

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	09/10/2026	DATE SUBMITTED:	09/14/2026
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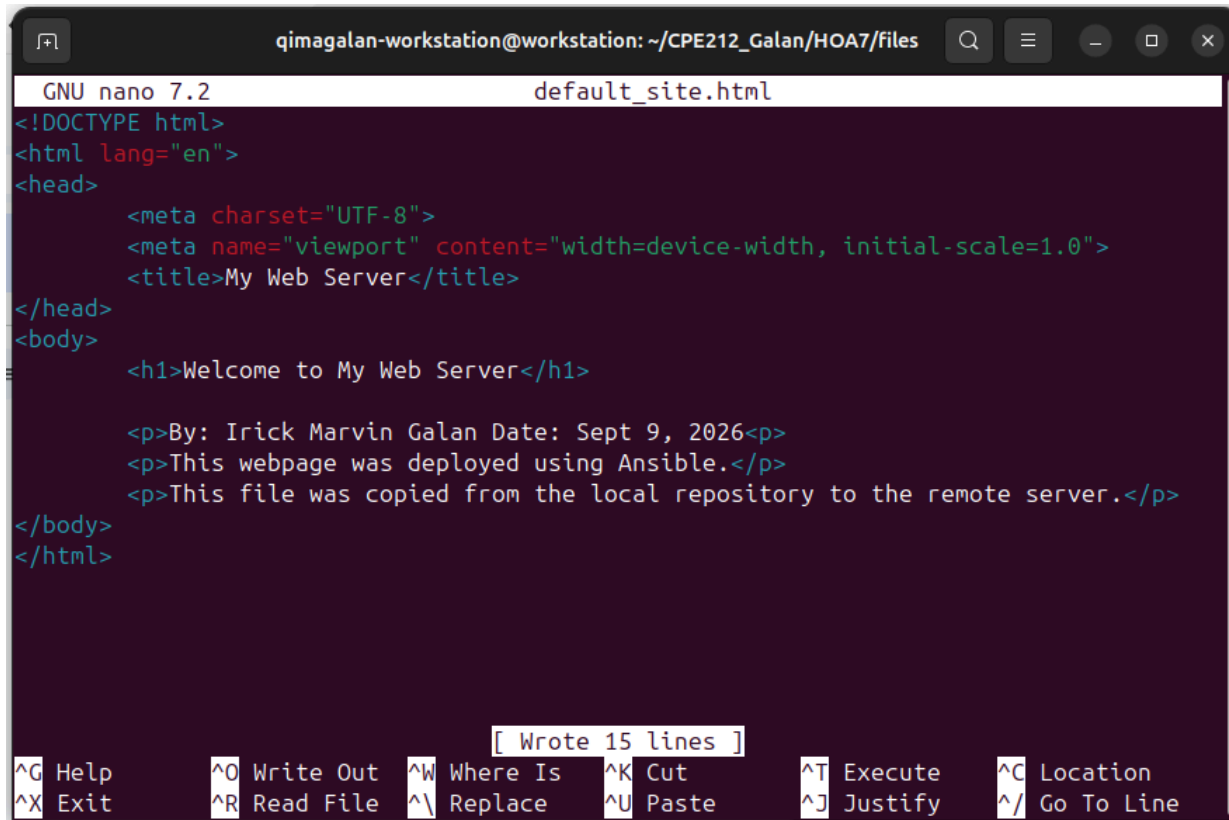
ACTIVITY NO. 7 Managing Files and Creating Roles in Ansible

1. PROCEDURE OUTPUT

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it "files."

Create a file inside that directory and name it "default_site.html." Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.



```

qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7/files
GNU nano 7.2 default_site.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Web Server</title>
</head>
<body>
  <h1>Welcome to My Web Server</h1>

  <p>By: Irick Marvin Galan Date: Sept 9, 2026</p>
  <p>This webpage was deployed using Ansible.</p>
  <p>This file was copied from the local repository to the remote server.</p>
</body>
</html>

[ Wrote 15 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
  
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
GNU nano 7.2 site.yml *
become: true
tasks:

- name: install apache and php for Ubuntu servers
  tags: apache, apache2, ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"

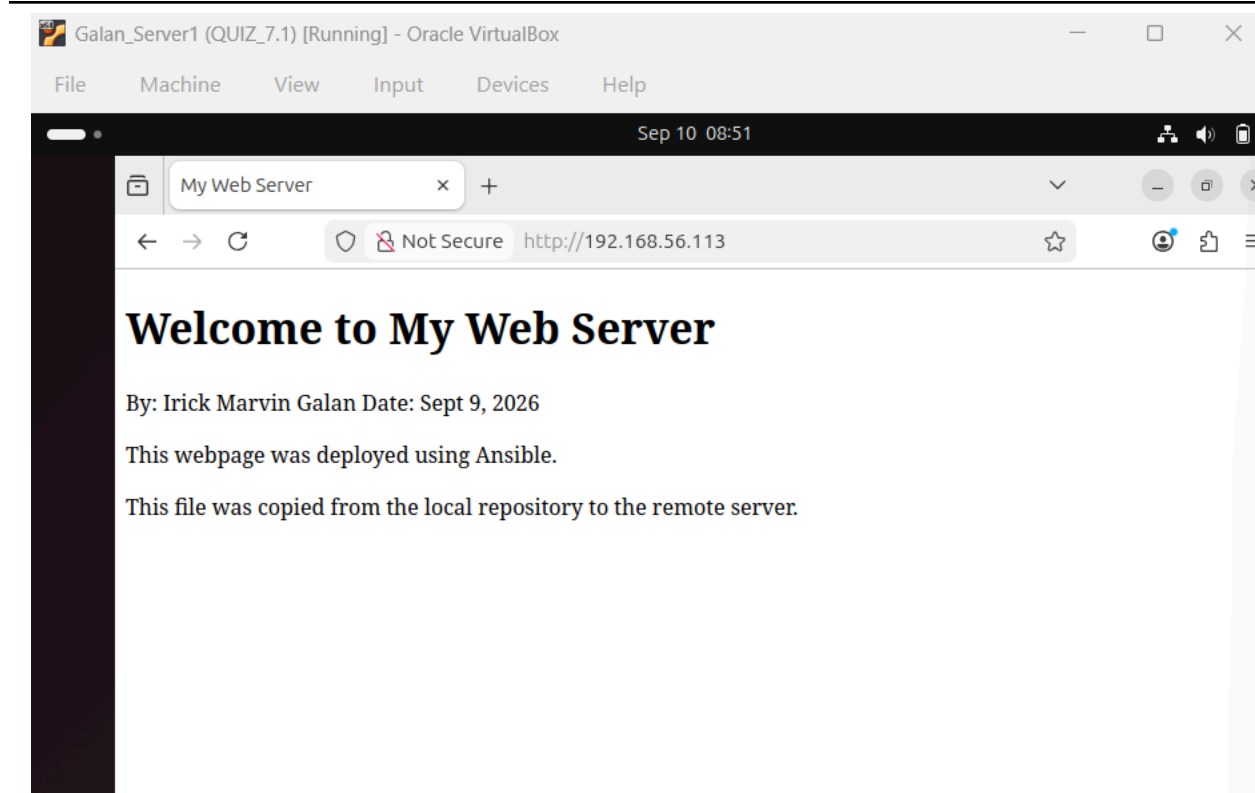
- name: copy default html for site
  tags: apache, apache2, httpd
  ansible.builtin.copy:
    src: "{{ ansible_env.HOME }}/CPE212_Galan/HOA7/files/default_site.html"
    dest: "/var/www/html/index.html"
    owner: root
    group: root
    mode: "0644"
```

3. Run the playbook site.yml. Describe the changes.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
PLAY [web_servers] *****
TASK [Gathering Facts] *****
ok: [192.168.56.113]
TASK [install apache and php for Ubuntu servers] *****
ok: [192.168.56.113]
TASK [install apache and php for CentOS servers] *****
skipping: [192.168.56.113]
TASK [copy default html for site] *****
changed: [192.168.56.113]
PLAY [file_servers] *****
```

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

The index.html under /var/www/html seems to be the default website tied to a machine's own IP Address. Therefore, when I entered the machine's IP in an ubuntu remote server, the contents of the copied html shows up.



5. Sync your local repository with GitHub and describe the changes.

```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
qimagalan-workstation@workstation:~/CPE212_Galan$ git branch
* home
  main
qimagalan-workstation@workstation:~/CPE212_Galan$ git add .
qimagalan-workstation@workstation:~/CPE212_Galan$ git commit -m "YAML for copying index.html to remote"
[home 6d465d8] YAML for copying index.html to remote
20 files changed, 344 insertions(+)
create mode 100644 HOA7/files/default_site.html
create mode 100644 HOA7/inventory.yml
create mode 100644 HOA7/role_playbook.yml
create mode 100644 HOA7/roles/dbservers_role/README.md
create mode 100644 HOA7/roles/dbservers_role/defaults/main.yml
create mode 100644 HOA7/roles/dbservers_role/handlers/main.yml
create mode 100644 HOA7/roles/dbservers_role/meta/main.yml
create mode 100644 HOA7/roles/dbservers_role/tasks/main.yml
create mode 100644 HOA7/roles/dbservers_role/tests/inventory
create mode 100644 HOA7/roles/dbservers_role/tests/test.yml
create mode 100644 HOA7/roles/dbservers_role/vars/main.yml
create mode 100644 HOA7/roles/webservers_role/README.md
create mode 100644 HOA7/roles/webservers_role/defaults/main.yml
create mode 100644 HOA7/roles/webservers_role/handlers/main.yml
create mode 100644 HOA7/roles/webservers_role/meta/main.yml
create mode 100644 HOA7/roles/webservers_role/tasks/main.yml
create mode 100644 HOA7/roles/webservers_role/tests/inventory
create mode 100644 HOA7/roles/webservers_role/tests/test.yml
create mode 100644 HOA7/roles/webservers_role/vars/main.yml
create mode 100644 HOA7/site.yml
qimagalan-workstation@workstation:~/CPE212_Galan$ git push origin home
Enumerating objects: 36, done.
Counting objects: 100% (36/36), done.
Delta compression using up to 4 threads
Compressing objects: 100% (18/18), done.
Writing objects: 100% (35/35), 4.62 KiB | 788.00 KiB/s, done.
Total 35 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:IrickMarvinGalan/CPE212_Galan.git
  9dd9b03..6d465d8  home -> home
qimagalan-workstation@workstation:~/CPE212_Galan$
```

Task 2: Download a file and extract it to a remote server

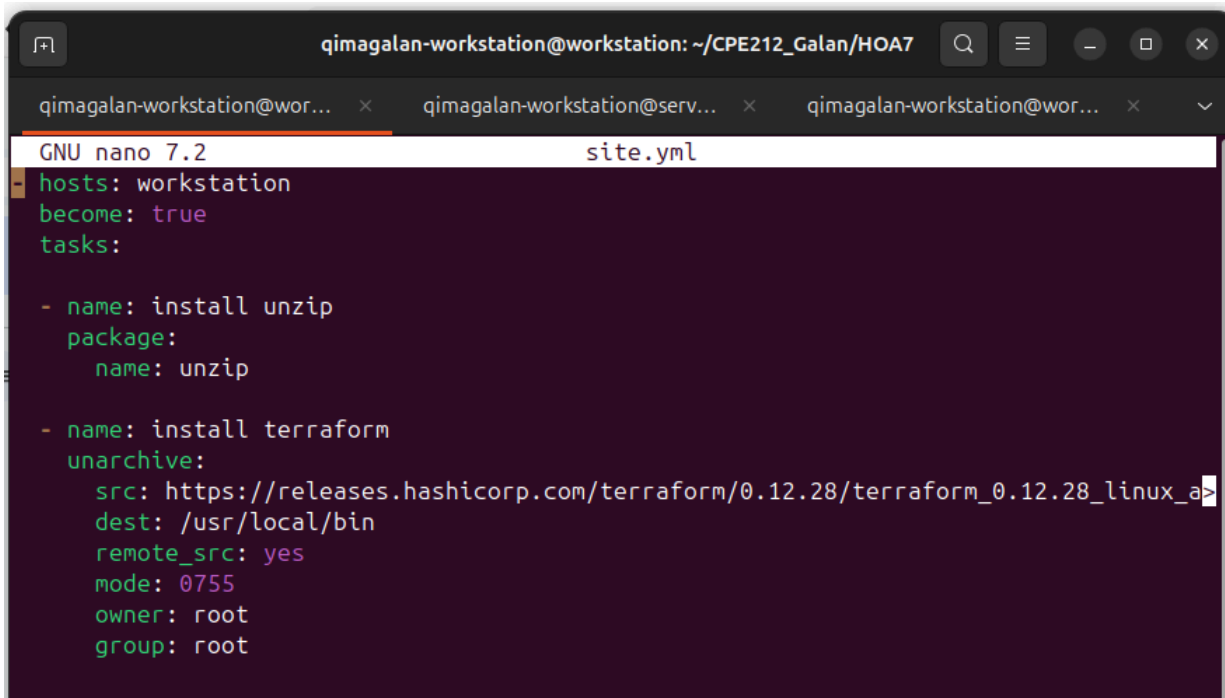
1. Edit the site.yml. Just before the web_servers play, create a new play:

- hosts: workstations

become: true

tasks:

```
- name: install unzip
package:
name: unzip
- name: install terraform
unarchive:
src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
dest: /usr/local/bin
remote_src: yes
mode: 0755
owner: root
group: root
note: In Ansible, a play is a section of a playbook that defines which remote machines Ansible should
manage and what tasks it should perform on them.
```



```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
GNU nano 7.2 site.yml
- hosts: workstation
  become: true
  tasks:

  - name: install unzip
    package:
      name: unzip

  - name: install terraform
    unarchive:
      src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_a>
```

2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
GNU nano 7.2 inventory.yml
[workstation]
192.168.56.114

[web_servers]
192.168.56.113

[db_servers]
192.168.56.106

[file_servers]
192.168.56.114 ansible_user=qimagalan
```

3. Run the playbook. Describe the output.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
PLAY [workstation] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]

TASK [install unzip] *****
ok: [192.168.56.114]

TASK [install terraform] *****
changed: [192.168.56.114]
```

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output. The .114 machine represents CentOS so I typed terraform in the CentOS machine, and it shows that the terraform is recognized and it outputs what command options we can use. Alternatively we can also use which terraform.

```
qimagalan@localhost:~  
[qimagalan@localhost ~]$ terraform  
Usage: terraform [-version] [-help] <command> [args]  
  
The available commands for execution are listed below.  
The most common, useful commands are shown first, followed by  
less common or more advanced commands. If you're just getting  
started with Terraform, stick with the common commands. For the  
other commands, please read the help and docs before usage.  
  
Common commands:  
  apply          Builds or changes infrastructure  
  console        Interactive console for Terraform interpolations  
  destroy        Destroy Terraform-managed infrastructure  
  env            Workspace management  
  fmt            Rewrites config files to canonical format  
  get            Download and install modules for the configuration  
  graph          Create a visual graph of Terraform resources  
  import         Import existing infrastructure into Terraform  
  init           Initialize a Terraform working directory  
  login          Obtain and save credentials for a remote host  
  logout         Remove locally-stored credentials for a remote host  
  output         Read an output from a state file  
  plan           Generate and show an execution plan  
  providers      Prints a tree of the providers used in the configuration
```

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)
Save the file and exit.


```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
GNU nano 7.2 site.yml
---
- hosts: all
  become: true
  pre_tasks:
    - name: update repository index (CentOS)
      tags: always
      dnf: update_cache: yes
      changed_when: false
      when: ansible_distribution == "CentOS"
    - name: install updates (Ubuntu)
      tags: always
      apt:
        update_cache: yes
        changed_when: false
      when: ansible_distribution == "Ubuntu"
- hosts: all
  become: true
  roles:
    - base
- hosts: workstation
  become: true
  roles:
    - workstations
```

```
- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7/roles
qimagalan-workstation@workstation:~/CPE212_Galan/HOA7$ cd roles
qimagalan-workstation@workstation:~/CPE212_Galan/HOA7/roles$ mkdir base web_servers fi
le_servers db_servers workstations
```

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7/roles
qimagalan-workstation@workstation:~/CPE212_Galan/HOA7/roles$ mkdir */tasks
mkdir: cannot create directory 'base/tasks': File exists
mkdir: cannot create directory 'db_servers/tasks': File exists
mkdir: cannot create directory 'file_servers/tasks': File exists
mkdir: cannot create directory 'web_servers/tasks': File exists
mkdir: cannot create directory 'workstations/tasks': File exists
```

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7/roles
qimagalan-workstation@wor... x qimagalan-workstation@serv... x qimagalan-workstation@wor... x
qimagalan-workstation@workstation:~/CPE212_Galan/HOA7/roles$ ls
base db_servers file_servers web_servers workstations
qimagalan-workstation@workstation:~/CPE212_Galan/HOA7/roles$ touch base/tasks/main.yml
db_servers/tasks/main.yml file_servers/tasks/main.yml web_servers/tasks/main.yml work
stations/tasks/main.yml
```

base Tasks:

```
GNU nano 7.2 roles/base/tasks/main.yml
---
- name: Prompt the Start of Play Execution
  ansible.builtin.debug:
    msg: "Executing the site.yml from Role: Base"
```

db_servers Tasks:

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
GNU nano 7.2 roles/db_servers/tasks/main.yml
---
- name: install mariadb package (CentOS)
  tags: centos, db, mariadb
  yum:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: install mariadb package (Ubuntu)
  tags: db, mariadb, ubuntu
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: "Mariadb- Restarting/Enabling"
  service:
    name: mysql
    state: restarted
    enabled: true
```

file_servers Tasks:

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@workstation: ~/CPE2... x qimagalan-workstation@workstation: ~/CPE2... x
GNU nano 7.2 roles/file_servers/tasks/main.yml
---
- name: start httpd
  tags: apache, centos, httpd
  service:
    name: httpd
    state: started
    enabled: true
  when: ansible_distribution == "CentOS"

- name: install samba package
  tags: samba
  package:
    name: samba
    state: latest
```

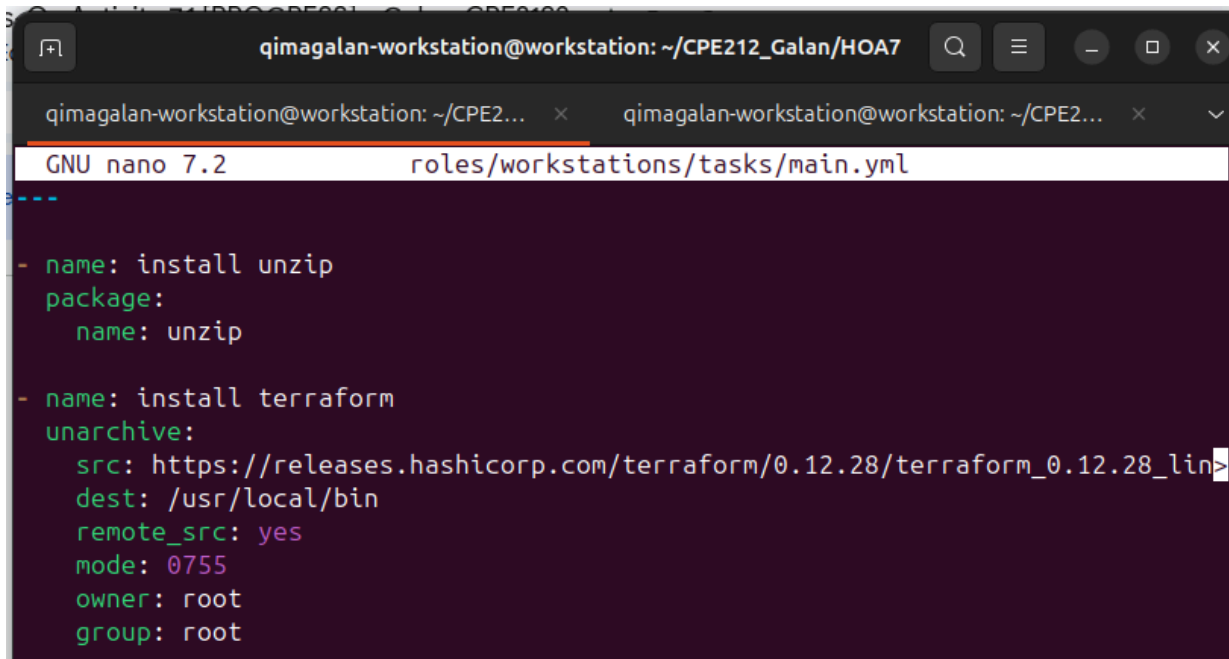
web_servers Tasks:

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@workstation: ~/CPE2... x qimagalan-workstation@workstation: ~/CPE2... x
GNU nano 7.2 roles/web_servers/tasks/main.yml
---
- name: install apache and php for Ubuntu servers
  tags: apache, apache2, ubuntu
  apt:
    name:
      - apache2
      - libapache2-mod-php
    state: latest
  when: ansible_distribution == "Ubuntu"

- name: install apache and php for CentOS servers
  tags: apache, centos, httpd
  dnf:
    name:
      - httpd
      - php
    state: latest
  when: ansible_distribution == "CentOS"
```

```
- name: copy default html for site
tags: apache,apache2,httpd
ansible.builtin.copy:
  src: "/home/qimagalan-workstation/CPE212_Galan/HOA7/files/default_site.html"
  dest: "/var/www/html/index.html"
  owner: root
  group: root
  mode: "0644"
```

workstations Tasks:



```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
GNU nano 7.2 roles/workstations/tasks/main.yml
---
- name: install unzip
  package:
    name: unzip

- name: install terraform
  unarchive:
    src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_lin>
    dest: /usr/local/bin
    remote_src: yes
    mode: 0755
    owner: root
    group: root
```

4. Run the site.yml playbook and describe the output.

By running the play that consolidates the roles, it is as if each /tasks/main.yml under each role is being run by the site.yml itself. It allowed the code to be modular while allowing the site yml to manage only the execution. This solves the problem of the code being difficult to maintain as the project grows larger.

```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7$ ansible-playbook -i inven
tory.yml site.yml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.106]
ok: [192.168.56.114]

TASK [update repository index (CentOS)] *****
skipping: [192.168.56.113]
skipping: [192.168.56.106]
ok: [192.168.56.114]

TASK [install updates (Ubuntu)] *****
skipping: [192.168.56.114]
ok: [192.168.56.113]
ok: [192.168.56.106]

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]
ok: [192.168.56.106]
ok: [192.168.56.114]
```



```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
TASK [base : Prompt the Start of Play Execution] *****
ok: [192.168.56.114] => {
  "msg": "Executing the site.yml from Role: Base"
}
ok: [192.168.56.113] => {
  "msg": "Executing the site.yml from Role: Base"
}
ok: [192.168.56.106] => {
  "msg": "Executing the site.yml from Role: Base"
}

PLAY [workstation] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]

TASK [workstations : install unzip] *****
ok: [192.168.56.114]

TASK [workstations : install terraform] *****
ok: [192.168.56.114]

PLAY [web_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.113]

TASK [web_servers : install apache and php for Ubuntu servers] *****
ok: [192.168.56.113]
```



```
qimagalan-workstation@workstation: ~/CPE212_Galan/HOA7
TASK [web_servers : install apache and php for CentOS servers] *****
skipping: [192.168.56.113]

TASK [web_servers : copy default html for site] *****
ok: [192.168.56.113]

PLAY [db_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.106]

TASK [db_servers : install mariadb package (CentOS)] *****
skipping: [192.168.56.106]

TASK [db_servers : install mariadb package (Ubuntu)] *****
ok: [192.168.56.106]

TASK [db_servers : Mariadb- Restarting/Enabling] *****
changed: [192.168.56.106]

PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]

TASK [file_servers : start httpd] *****
ok: [192.168.56.114]

TASK [file_servers : install samba package] *****
```

```
PLAY [file_servers] *****

TASK [Gathering Facts] *****
ok: [192.168.56.114]

TASK [file_servers : start httpd] *****
ok: [192.168.56.114]

TASK [file_servers : install samba package] *****
ok: [192.168.56.114]

PLAY RECAP *****
192.168.56.106      : ok=7    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.113      : ok=7    changed=0    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.114      : ok=10   changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0

qimagalan-workstation@workstation:~/CPE212_Galan/H0A7$
```

Push to Github:

```
qimagalan-workstation@workstation:~/CPE212_Galan$ git push origin home
Enumerating objects: 27, done.
Counting objects: 100% (27/27), done.
Delta compression using up to 4 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (22/22), 2.68 KiB | 392.00 KiB/s, done.
Total 22 (delta 6), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (6/6), completed with 2 local objects.
To github.com:IrickMarvinGalan/CPE212_Galan.git
6d465d8..cfdd858  home -> home
qimagalan-workstation@workstation:~/CPE212_Galan$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?

By creating roles we are effectively grouping related code together instead of every process cramped into a single file. Maintenance and debugging is made simpler if we are only skimming through relevant pieces of code. From the previous activity we also noticed how there are vars directory in each role created by ansible-galaxy, and this allows proper segregation of defined variables preventing other roles from misusing the value intended for other roles.

2. What is the importance of managing files?

Linux distributions make use of file systems to manage every single component of the operating systems which makes remote file management skills in ansible a requirement. Remote servers and clients need to be reconfigured

from time to time, configurations and settings copied and imported, and etc.

Conclusions/Learnings:

In this activity, we explored two essential ansible skills which include file management, creating and utilizing roles for codebase modularization. I learned how `roles/main.yml` should contain only tasks and their names, meanwhile the `main.yml` is the one to indicate the target metagroup and the execution of roles.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.